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Clinical Sports Medicine Update

Use of Publicly Obtained Data in Sports Medicine Research

A Systematic Review and Bibliometric Analysis

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Background: Publicly obtained data (POD) have recently been utilized frequently by sports medicine researchers to describe injury patterns, risk factors, and outcomes in elite athletes. The relative ease of this type of research that is based solely on internet and media sources has resulted in a near exponential increase in the number of these POD studies.

Purpose: To systematically review the sports medicine literature for studies based solely on POD.

Study Design: Systematic review and bibliometric analysis; Level of evidence, 4.

Methods: A systematic review of POD studies published since 2000 was conducted in accordance with the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines. Studies of interest were those relying on publicly available injury reports or online media for data acquisition in collegiate, semiprofessional, and professional athletes.

Results: There were 209 POD studies published between 2000 and 2022, with 173 (82.8%) of these studies published after 2016. Studies were published most frequently on athletes participating in North American professional leagues: National Football League (n = 69 [28.4%]), Major League Baseball (n = 56 [23.0%]), National Basketball Association (n = 37 [15.2%]), and National Hockey League (n = 33 [13.6%]). The most common injuries assessed were head injuries/concussions (n = 43 [21.1%]), anterior cruciate ligament injuries (n = 33 [16.2%]), and ulnar collateral ligament injuries (n = 23 [11.3%]). One-quarter of the studies (n = 53 [25.4%]) reported only 1 POD source, and 1 study (0.5%) reported no source. Additionally, 65 studies (31.1%) listed nonspecific POD resources or solely cited previous literature to describe the POD search methodology and data acquisition.

Conclusion: POD studies are exponentially increasing in number, particularly across major North American professional sports leagues, with significant variability in the injury of interest, search methodology, and number of data sources. The accuracy of the conclusions reached based on the POD methodology appears highly variable. Given the potential impact of these publications as both contributors to current knowledge and drivers of future research, the sports medicine community should be aware of the inherent biases and limitations of POD injury studies.

Keywords: sports medicine epidemiology; publicly obtained data; POD; professional athletics

Epidemiological research in the field of sports medicine seeks to define an athlete's risk for an injury,[24] identify modifiable and nonmodifiable risk factors for such an injury,[2] and provide prognostic data regarding an athlete's ability to return to competition.[18,27] Frequently, this type of research utilizes well-curated prospective data collected by major sports leagues including the National Football League (NFL),[4] the National Basketball Association (NBA),[13] the National Hockey League (NHL),[19] Major League Soccer (MLS),[1] Major League Baseball (MLB),[3] and the National Collegiate Athletic Association (NCAA).[28] Research provided by these surveillance databases has significantly informed the sports medicine community's understanding of the sport-specific injury risk, injury patterns,[24] and implementation of rule changes and other preventive measures, which has translated into improvements in both professional and amateur athlete health and safety.[14]

Recently, however, researchers have turned to readily available online injury reports, media coverage, and player profiles to collate data for the purposes of scientific inquiry.[8] The primary impetus for the use of these publicly obtained data (POD) appears to be the relative difficulty in acquiring access to various league-sanctioned databases by researchers without league affiliation. Recent studies

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evaluating publicly available injury data in NFL athletes (ie, anterior cruciate ligament [ACL] injuries and concussions) suggest that this method of research may fail to identify a significant number of less severe injuries (thereby overestimating the injury severity) and may lead to bias in capturing injuries only in higher profile athletes (thereby underestimating the injury incidence).[8]

Given the perceived increased prevalence of POD studies in sports medicine research, and the concurrent concerns associated with this type of research, the purpose of this study was to systematically review studies on major North American professional, semiprofessional, and collegiate sports leagues that rely solely on POD sources.

METHODS

Information Sources

This systematic review was conducted in accordance with the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines.[20] The published literature was queried utilizing search strategies devised by a medical librarian for POD studies in sports medicine research. The search strategies used a combination of standardized terms and keywords including, but not limited to (sports injuries OR athletic injuries) AND (professional OR national association OR national federation OR sports organization OR football club OR major league OR minor league OR semi-professional OR collegiate OR NCAA) AND (public data OR retrospective OR cohort study) (see Appendix). The search was performed in February 2022 in the Ovid MEDLINE, Embase, Web of Science, and CINAHL databases, with date limits of 2000 to 2022.

Eligibility Criteria

Articles eligible for inclusion were POD studies in sports medicine performed over the past 2 decades (2000-2022). Specifically, studies were included if they utilized publicly available injury reports in print or online media and were composed of athletes at the professional, semiprofessional, or collegiate level.

Selection Process

There were 2 authors (P.M.I., A.W.K.) who independently screened and reviewed the full text of retrieved studies meeting the inclusion/exclusion criteria. Studies were excluded if they (1) were abstracts, conference proceedings, letters, reviews, or editorials; (2) were strictly video analyses; (3) included official league electronic health records or injury reports; and (4) did not analyze a cohort of athletes. For studies that the 2 reviewing authors disagreed on, a third author (M.J.M) acted as the arbitrator, if needed. A kappa coefficient was calculated to assess interrater reliability agreement on inclusion.

Data Elements

Again, 2 authors (P.M.I., A.W.K.) worked independently to extract data from the selected studies. If there were any missing data points, these 2 authors attempted to contact the study authors via email. Bibliometric data extracted from each study included the full list of authors, year of publication, journal, institution of investigation, study design, level of evidence, sample/league (ie, NFL, NHL, MLB, NCAA, etc), sample size (total number of athletes/injuries analyzed), data sources utilized for injuries, anatomic region of interest, specific injury of interest (eg, ACL rupture, concussion), and outcomes of interest (eg, player performance, return to play). Articles that focused on a broad collection of injuries (eg, any injury resulting in time lost, any injury recorded on an injury report, any injury resulting in an injured reserve or disabled list transaction) were recorded under the "all injuries" designation. The journal impact factor (IF) was determined through Journal Citation Reports (Clarivate). No formal assessment of bias was performed. Data were synthesized, and summary statistics were reported including counts and percentages, means and standard deviations, and medians and interquartile ranges, where appropriate. All analyses were conducted at the 95% CI level with the statistical software program SPSS (Version 27.0; IBM).

RESULTS

Study Selection

From the initial database searches, a total of 11,537 unique citations were returned. Of these, 5947 citations were duplicates that were removed, resulting in 5590 studies. Then, 245 studies met the inclusion criteria for a full-text review. After reviewing the studies, 2 raters (A.W.K., S.C.T.) reached 93.5% agreement, with the Cohen kappa demonstrating "substantial" agreement (kappa = 0.72). After a deliberation between these 2 authors, 100.0%

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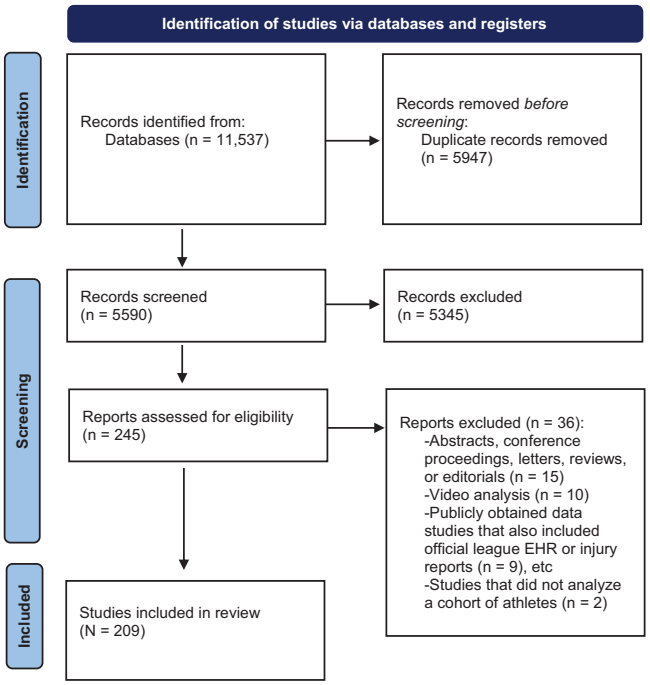

Figure 1. PRISMA[24] (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) flowchart for study inclusion. EHR, electronic health record.

agreement on inclusion/exclusion was obtained; ultimately, 209 studies were included for final analysis (Figure 1).[23]

Study Characteristics

There were 209 POD studies published between 2000 and 2022, with 173 (82.8%) published after 2016 (Figure 2; also see Appendix Table A1, available online).

Overall, 179 studies (85.6%) were published at institutions within the United States. There were 57 unique journals that published POD studies, with the most studies published in *The Orthopaedic Journal of Sports Medicine* (n = 62 [29.7%]; IF = 3.4), *The American Journal of Sports Medicine* (n = 25 [12.0%]; IF = 7.1), *The Physician and Sportsmedicine* (n = 13 [6.2%]; IF = 2.76), *Arthroscopy, Sports Medicine, and Rehabilitation* (n = 10 [4.8%]; nonindexed), and *Journal of Shoulder and Elbow Surgery* (n = 9 [4.3%]; IF = 3.51). A complete list of journals, with associated IFs and open access designations, can be found in Appendix Table A2 (available online). Further, 95 studies (45.5%) were published in open access journals. The majority of articles were either retrospective cohort (n = 116 [55.5%]) or case-control (n = 53 [25.4%]) studies, providing mostly level 3 or 4 evidence.[26] The mean methodological index for non-randomized studies (MINORS) score was 10.95 for noncomparative studies and 18.58 for comparative studies (Appendix Table A2, available online).

There were 198 studies (94.7%) that examined a single sport, whereas 11 studies (5.3%) included >1 sport. Sports commonly studied were football (n = 71 [29.1%]), baseball

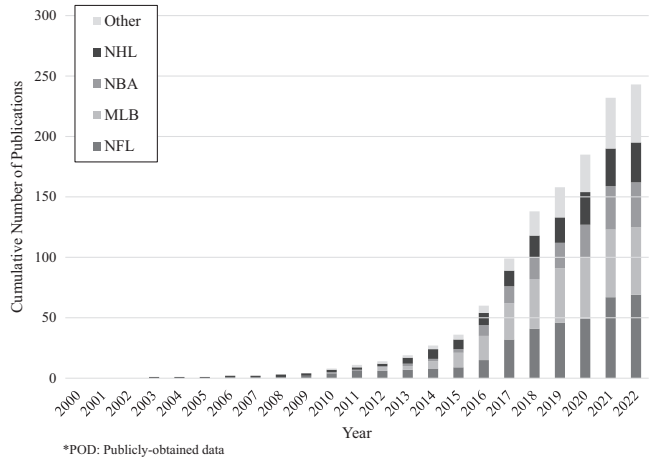

Figure 2. The overall cumulative frequency of publicly obtained data (POD) studies published between 2000 and 2022 delineated by league. MLB, Major League Baseball; NBA, National Basketball League; NFL, National Football League; NHL, National Hockey League.

(n = 58 [23.8%]), basketball (n = 44 [18.0%]), ice hockey (n = 33 [13.5%]), and soccer (n = 22 [9.0%]). Leagues more frequently analyzed were the NFL (n = 69 [28.4%]), MLB (n = 56 [23.0%]), the NBA (n = 37 [15.2%]), and the NHL (n = 33 [13.6%]). The most common anatomic regions of interest were the head (n = 47 [19.2%]), all regions (n = 45 [18.4%]), and the knee (n = 37 [15.1%]). Specific injuries most frequently analyzed were head injuries/concussions (n = 43 [21.1%]), ACL injuries (n = 33 [16.2%]), ulnar collateral ligament injuries (n = 23 [11.3%]), and Achilles tendon injuries (n = 22 [10.8%]). Complete descriptive data can be found in Table 1.

In the NFL (n = 16 [25.0%]) and NHL (n = 11 [40.7%]), head injuries/concussions were collectively the most common injuries studied. Head injuries/concussions were also studied frequently in MLB (n = 6 [12.2%]) and the NBA (n = 4 [11.8%]). In the NFL and NBA, lower extremity injuries, including ACL and Achilles tendon injuries, were frequently studied compared with MLB in which ulnar collateral ligament injuries were the most analyzed (Table 2).

Researchers utilized POD to assess player performance most frequently (n = 109 [28.2%]), followed by return to play (n = 85 [22.0%]) and injury incidence (n = 80 [20.7%]) (Figure 3).

Most studies (n = 124 [59.3%]) reported ≥3 sources that they utilized for data collection; 31 (14.8%) reported 2 sources, 53 (25.4%) reported 1 source, and 1 study (0.5%) reported no sources. In addition, 65 studies (31.1%) listed nondescript public data resources (ie, "team websites," "player websites," "online media," "press releases") or simply referred to previous literature.

DISCUSSION

The utilization of POD injury methodology in sports medicine has increased exponentially in the past decade, with

TABLE 1
Characteristics of Included Studies[a]

	n (%)
Country of origin	
United States	180 (86.1)
Germany	10 (4.8)
Canada	9 (4.3)
Australia	6 (2.9)
United Kingdom	2 (1.0)
Italy	1 (0.5)
Brazil	1 (0.5)
Study design	
Retrospective cohort	116 (55.5)
Case-control	53 (25.4)
Cross-sectional	22 (10.5)
Case series	16 (7.7)
Prospective cohort	2 (1.0)
Sport	
Football	71 (29.1)
Baseball	58 (23.8)
Basketball	44 (18.0)
Ice hockey	33 (13.5%)
Soccer	22 (9.0)
Australian football	5 (2.0)
Martial arts	2 (0.8)
Other[b]	9 (3.7)
League	
NFL	69 (28.4)
MLB	56 (23.0)
NBA	37 (15.2)
NHL	33 (13.6)
UEFA	18 (7.4)
MLS	5 (2.1)
NCAA	5 (2.1)
WNBA	5 (2.1)
AFL	4 (1.6)
Bundesliga	4 (1.6)
Other[c]	7 (2.9)
Anatomic region of interest	
Head	47 (19.2)
All	45 (18.4)
Knee	37 (15.1)
Elbow	28 (11.4)
Ankle	24 (9.8)
Shoulder	16 (6.5)
Neck/spine	12 (4.9)
Hip	8 (3.3)
Abdomen	5 (2.0)
Hand	5 (2.0)
Wrist	4 (1.6)
Foot	3 (1.2)
Fatality	2 (0.8)
Lower extremity	2 (0.8)
Other[d]	7 (2.9)
Specific injury	
Head injury/concussion	43 (21.3)
ACL injury	33 (16.3)
Elbow ulnar collateral ligament injury	23 (11.4)
Achilles tendon injury	22 (10.9)
Craniofacial injury	8 (4.0)
Cervical spine injury	7 (3.5)
Lumbar spine injury	7 (3.5)
Shoulder instability	7 (3.5)
All other fractures	6 (3.0)
Intra-articular hip injury (arthroscopic surgery)	5 (2.5)
Knee cartilage injury (microfracture)	5 (2.5)
Deep vein thrombosis/pulmonary embolism	4 (2.0)
Patellar tendon injury	3 (1.5)
Pectoralis major rupture	3 (1.5)
Adductor injury	2 (1.0)
Clavicular fracture	2 (1.0)
Meniscal tear	2 (1.0)
Other[e]	20 (10.0)

[a]ACL, anterior cruciate ligament; AFL, Australian Football League; MLB, Major League Baseball; MLS, Major League Soccer; NBA, National Basketball Association; NCAA, National Collegiate Athletic Association; NFL, National Football League; NHL, National Hockey League; UEFA, Union of European Football Associations; WNBA, Women's National Basketball Association.

[b]Automobile racing, boxing, cricket, cycling, Gaelic football, hurling, golf, rugby, skiing (n = 1 each).

[c]Gaelic Athletic Association, International Mixed Martial Arts Federation, Ontario Hockey League, Professional Golfers' Association, Professional Rugby, Union Cycliste Internationale (Tour de France), Ultimate Fighting Championship (n = 1 each).

[d]Cardiovascular, forearm, illness, mortality, neuromuscular, size (BMI), pitch volume (n = 1 each).

[e]Abdominal muscle injury, acromioclavicular joint injury, all shoulder injuries, amyotrophic lateral sclerosis, arm fatigue, distal biceps injury, hamstring injury, high ankle sprain, hook of hamate injury, Jones fracture, shoulder labral injury, lower extremity injury, medial collateral ligament injury, metacarpal fracture, multiligament knee injury, thoracic outlet syndrome, thumb ulnar collateral ligament injury, upper extremity injury, wrist fracture (n = 1 each).

209 studies published since 2000. The majority (>80%) of these studies were published in the past 5 years, many of which are in high-impact journals (ie, *The American Journal of Sports Medicine*, *Arthroscopy*, *Foot & Ankle International*, *JAMA Network Open*, *Spine*, and *Journal of Shoulder and Elbow Surgery*) (see Appendix Table A2 [available online] for IF details and subscription versus open access status). These articles covered nearly all major North American professional sports leagues, utilized a myriad of online sources for data procurement, and reflected a variety of different injuries of interest. Unfortunately, these studies varied significantly in methodology and in the detail with which this methodology was described. Given the impact of these publications as both contributors to current knowledge and drivers of future research, the sports medicine community should thoroughly assess the assumed benefits and inherent limitations and concerns associated with this method of study.[11,14]

One potential explanation for the exponential increase in these studies is the apparent advantages of this method of research: the ability to construct a sport- or league-based injury data set with minimal cost, the ability to leverage

TABLE 2
Specific Injuries in Studies by League[a]

	n (%)
NFL	64 (100.0)
Head injury/concussion	16 (25.0)
ACL injury	13 (20.3)
Achilles tendon injury	5 (7.8)
Cervical spine injury	4 (6.3)
Lumbar spine injury	4 (6.3)
Pectoralis major rupture	3 (4.7)
Other	19 (29.7)
MLB	49 (100.0)
Ulnar collateral ligament injury	22 (44.9)
Head injury/concussion	6 (12.2)
Achilles tendon injury	3 (6.1)
Other	18 (36.7)
NBA	34 (100.0)
ACL injury	8 (23.5)
Achilles tendon injury	5 (14.7)
Head injury/concussion	4 (11.8)
Shoulder instability	3 (8.8)
Other	14 (41.2)
NHL	27 (100.0)
Head injury/concussion	11 (40.7)
Craniofacial injury	4 (14.8)
ACL injury	3 (11.1)
Shoulder instability	2 (7.4)
Other	7 (25.9)

[a]ACL, anterior cruciate ligament; MLB, Major League Baseball; NBA, National Basketball Association; NFL, National Football League; NHL, National Hockey League.

young researchers (ie, medical students and residents) to construct data sets, the ability to bypass oversight required for data obtained under a collectively bargained agreement between a league and its players,[11] the ability for relatively expeditious article preparation once a data set has been established, and the ability to publish numerous articles from a single systematic search or source.[7,15-17,29,30,32,33,39] Despite these advantages, this method of research possesses intrinsic disadvantages and limitations.[10] Specifically, nonstandardized data sources may lead to the utilization of incomplete, inaccurate, or unclear data that may result in errant conclusions, thereby providing misinformation on an injury's relative influence, severity, treatment, and ultimate prognosis.[10] Additionally, outcomes from this type of research are inclined to demonstrate statistical performance[9,34,38,41] or financial loss,[21,31] which can be affected by numerous factors beyond injury characteristics, treatment, and/or physiological recovery. The recent exponential increase in publications utilizing POD methodology across all North American professional sports leagues emphasizes the need for critical analysis of this method of scientific inquiry.

When evaluating methodological scores (ie, MINORS scores) for these articles, nearly all publications share 2 shortcomings: all are retrospective in nature and lack preoperative sample size estimates. Given that these publications rely on previously published news reports, the

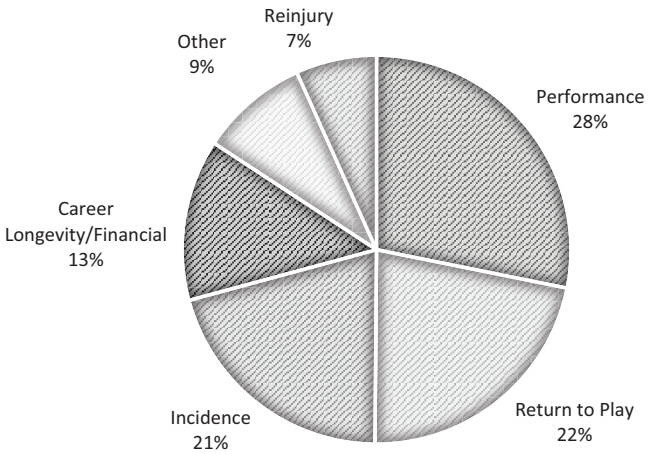

Figure 3. Outcomes most frequently analyzed in publicly obtained data (POD) studies from 2000 to 2022.

retrospective nature of POD is inherent to the methodology. However, we also noticed many factors that may uniquely affect the accuracy and quality of each specific article: data source utilized, reproducibility of search strategies, number of reviewers, cross-reference of identified injuries, specificity of injury reports available, and time frame selected. For example, numerous studies utilize date ranges that predate the internet or electronic records, presumably drastically increasing the variability and accuracy of that study. While we have provided methodological scoring within this article, future authors may consider POD-specific methodological scoring to better differentiate between studies. However, we believe that even methodologically rigorous POD studies face inherent limitations that cannot be overcome without prospectively collected and curated data.

The impact of these studies on the scientific literature is not unique to a single sport or league but rather has implications across all professional and amateur athletics. Although this study does not purport to objectively evaluate the accuracy (ie, "capture rate") of the included articles, the increasing utilization of this methodology warrants critical analysis and validation by the sports medicine community. For example, head injuries/concussions were a consistent area of interest across all sports, although injury analysis varied by sport. Significant heterogeneity existed among data sources utilized for each study, and a subset of POD studies did not provide significant detail regarding the search methodology or specific data source. Although we evaluated the methodology of this study criterion in relation to other descriptive studies, such scores may fail to fully encapsulate or distinguish between studies using POD. Rather, an opportunity may exist to define "best practices" or a methodological scoring system for this unique class of studies.

Importantly, although discerning researchers and clinicians may be able to fairly interpret these articles with consideration of potential limitations, this nuanced approach to interpretation is an unfair and inappropriate expectation of the media and public. Yet, given the relative frequency of POD studies, the proportion of articles

published in high-impact journals, and the proclivity for publication in open access journals, the impact of these studies may extend beyond the sports medicine community. Numerous articles identified within the systematic review have been discussed and publicized by media outlets: ESPN,[19,22] CBS Sports,[35,37] *USA Today*,[6,40] The Athletic,[25,36] and fan base–specific team websites.[5,19] As such, POD studies may drive unrealistic expectations for the relevance of these studies by people outside of the sports medicine community.

Although the composition of studies by injury type varied across leagues, concussions and head/neurological injuries represented a significant proportion of the existing POD injury literature. The diagnosis of a head injury and concussion, even by trained and experienced sideline physicians performing an examination and objective testing, is a challenge for any form of clinical care or research.[12] This difficulty may partially account for the significant percentage of missed injuries noted in POD studies evaluating NFL concussions. These challenges, and related shortcomings, may be more prominent in other leagues because of the variability in league-specific guidelines. Given the media and public's attention regarding this subject, unclear or inaccurate studies may negatively affect the career and financial potential of these professional athletes, rule changes, and public trust in medical research and in the transparency of the specific leagues for their injury data. Therefore, particular attention must be paid to evaluate this body of literature and understand how the quality of research may differ across leagues or data sources.

Studies also varied significantly in the number of sources utilized and the detail with which these sources were described. Many sources referred simply to previously utilized methodology or only used generalities to describe data acquisition. This variability in source utilization likely leads to significant heterogeneity in data capture. In fact, previous work demonstrated that POD methodology allowed certain articles to capture >90% of all injuries, while others captured <30%[8] (Appendix, available online). As such, the standardization of both research methodology and reporting method may result in more reliable data capture. Additionally, one of the primary roles of epidemiological research is to assess the impact of rule changes or conditions on athlete health. Given the current heterogeneity of methods and possible inability to accurately reproduce studies based on described search strategies, POD methodology, as currently structured, may be unable to differentiate changes in the injury incidence or severity from artifacts related to study design and search strategies.

This systematic review and bibliometric analysis does possess limitations, primarily related to the definition of POD methodology. For example, select studies utilized public data in combination with video analysis or an injury surveillance data set. However, interrater reliability remained high, and most studies relied entirely on POD methodology. Despite our concerns regarding the accuracy of POD studies, we did not seek to evaluate the accuracy or validity of this method of research based on the articles that were reviewed. However, future research may build on the results of the current study as well as previous published literature to clarify what role, if any, studies based solely on public data has in the care of the injured athlete.

CONCLUSION

POD studies are exponentially increasing in number, particularly across major North American professional sports leagues, with significant variability in the injury of interest, search methodology, and number of data sources. The accuracy of the conclusions reached based on the POD methodology appears highly variable. Given the potential impact of these publications as both contributors to current knowledge and drivers of future research, the sports medicine community should be aware of the inherent biases and limitations of POD injury studies.

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